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Nicholas M. Rapidis

Curriculum Vitae

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Academic Positions

2026- **Yale University**
Yale Quantum Institute Fellow
Postdoctoral researcher in Prof. Konrad Lehnert's group developing superconducting quantum sensors for readout of electromagnetic systems, such as dark matter axion experiments.

Education

2019-2026 **Stanford University**
Ph.D. in Physics
Thesis Title: *Sub-μeV Dark Matter Axion Searches: Design and Realization* [Stanford Libraries]
Advisor: *Prof. Kent Irwin*
M.S. in Physics completed in July 2022

2015-2019 **University of California, Berkeley**
B.A. in Physics. Graduated with High Distinction in General Scholarship and Departmental Honors
Thesis Title: *Resonant Axion-Photon Scattering and Galactic Searches for Axions*
Advisor: *Prof. Karl van Bibber*

Research Experience

2020-present **Graduate Research Assistant**, Stanford University
Advisor: *Prof. Kent Irwin*
Member of the Dark Matter Radio (DMRadio) collaboration. Presented DMRadio-Core. Presented design concept and first models of DMRadio-GUT. Lead on the design, modeling, and optimization of science reach of DMRadio-m^3. Lead on DMRadio-50L SQUID readout chain, superconducting sheath design, and contributions to other cryogenic hardware components. Former lead on SQUID testing for BICEP array.

2019-2020 **Graduate Research Assistant**, Stanford Institute for Theoretical Physics
Advisor: *Prof. Savas Dimopoulos*
Studied phenomenology of dense dark matter axion clumps (oscillons) collisions with neutron stars. Set limits on oscillon dark matter abundance.

2016-2019 **Undergraduate Research Assistant**, UC Berkeley

Advisor: *Prof. Karl van Bibber*

Member of the *Haloscope at Yale Sensitive to Axion Cold Dark Matter (HAYSTAC)* collaboration. Co-lead on refurbishment and optimization for cavity used HAYSTAC Phase II. Introduced extensive use of finite element simulation techniques for axion cavity characterization.

Honors & Awards

2026 *Springer Thesis Award*

2022 *Young Scientist Award* at *Identification of Dark Matter 2022*: One of best three talks (out of 90) given by graduate students and postdocs at the conference.

2019 *Phi Beta Kappa*

2018-2019 UC Berkeley *Haas Scholar*

Professional Activities

2023- Journal Referee: *Physical Review Letters*, *Physical Review D*, *Journal of Low Temperature of Physics*

Publications

[INSPIRE Profile] [Google Scholar Profile]

Principal Author

[19] **DMRadio-Core: A new approach to GUT-scale axion searches**
V. Ankel *et al.* *Phys. Rev. D* **114**, 092004 (2026) [arXiv:2604.16602][INSPIRE]

[18] **Electromagnetic modeling and science reach of DMRadio-m^3**
A. AlShirawi *et al.* *Phys. Rev. D* **112**, 052001 (2025) [arXiv:2302.14084][INSPIRE]

[17] **Characterization of the HAYSTAC axion dark matter search cavity using microwave measurement and simulation techniques**
N.M. Rapidis *et al.* *Review of Scientific Instruments* **90**, 024706 (2019) [arXiv:1809.02246][INSPIRE].

Joint Principal Author

[16] **Resonant Conversion of Dark Matter Oscillons in Pulsar Magnetospheres**
A. Prabhu and N.M. Rapidis, *JCAP* **10**, (2020) 054 [arXiv:2005.03700][INSPIRE].

Contributing Author

[15] **The Cryogenic System of DMRadio-50L**
V. Ankel *et al.* [arXiv:2607.05771] [INSPIRE]

[14] **Search for Dark Photons between 16.96 – 19.52 μeV with the HAYSTAC experiment**
X. Bai *et al.* *Phys. Rev. D* **113**, 012010 (2026) [arXiv:2510.15848] [INSPIRE]

[13] **Noise limits for dc SQUID readout of high-Q resonators below 300 MHz**
V. Ankel *et al.* *J. Appl. Phys.* **138**, 094505 (2025) [arXiv:2504.20398] [INSPIRE]

[12] **Dark Matter Axion Search with HAYSTAC Phase II**
X. Bai *et al.* *Phys. Rev. Lett.* **134**, 151006 (2025) [arXiv:2409.08998] [INSPIRE]

[11] **Measurements of DC SQUID Damping Effects on Superconducting Resonant Circuits**
E. C. van Assendelft *et al.* *IEEE Transactions on Applied Superconductivity* **33**, 5, (2023) [INSPIRE]

[10] **New Results from HAYSTAC's Phase II Operation with a Squeezed State Receiver**
M.J. Jewell *et al.* *Phys. Rev. D* **107**, 072007, (2023) [arXiv:2301.09721][INSPIRE]

[9] **Quantum metrology of low frequency electromagnetic modes with frequency upconverters**
S.E. Kuenstner *et al.* Phys. Rev. Research **7**, 013281 (2025)[arXiv:2210.05576][INSPIRE]

[8] **DMRadio-m^3: A Search for the QCD Axion Below 1 μeV**
L. Brouwer *et al.* *Phys. Rev. D* **106**, 103008, (2022) [arXiv:2204.13781][INSPIRE]

[7] **Introducing DMRadio-GUT, a search for GUT-scale QCD axions**
L. Brouwer *et al.* *Phys. Rev. D* **106**, 112003, (2022) [arXiv:2203.11246][INSPIRE]

[6] **A Model-Independent Radio Telescope Dark Matter Search**
A. Keller, *et al.* *Astrophys. J.* **927** (2022) 1, 71. [arXiv:2112.03439][INSPIRE]

[5] **A quantum-enhanced search for dark matter axions**
K.M. Backes *et al.* *Nature* **590**, 238-242 (2021) [arXiv:2008.01853][INSPIRE]

[4] **An improved analysis framework for axion dark matter searches**
D.A. Palken *et al.* *Phys. Rev. D* **101**, 123011, (2020) [arXiv:2003.08510][INSPIRE].

[3] **Results from Phase 1 of the HAYSTAC microwave cavity axion experiment**
L. Zhong *et al.* *Phys. Rev. D* **97**, 092001, (2018) [arXiv:1803.03690][INSPIRE].

[2] **Design and Operational Experience of a Microwave Cavity Axion Detector for the 20-100 μeV Range**
S. Al Kenany *et al.* *Nuclear Instruments and Methods in Physics Research A* **854** (2017) 11–24. [arXiv:1611.07123] [INSPIRE].

[1] **First Results from a Microwave Cavity Axion Search at 24 μeV**
B.M. Brubaker *et al.* *Phys. Rev. Lett.* **118**, 061302 (2017) [arXiv:1610.02580][INSPIRE].

Talks

Invited Talks

Searching for sub-μeV axions with DMRadio
Oak Ridge National Lab, June 4, 2026.

GUT scale axion searches with DMRadio
SLAC National Accelerator Laboratory, Elementary Particle Physics Theory Seminar, Dec. 10, 2025

Searching for sub-μeV axions below the standard quantum limit
Yale University, Yale Quantum Institute Seminar, December 1, 2025

Searching for low mass dark matter axions with DMRadio
Boston University, Physics Seminar, July 24, 2024

Status of the DMRadio Program
YOUNGST@RS - Shoot for the Stars, Aim for the Axions, October 4-7, 2022, Virtual.

Selected Contributed Talks

DMRadio-GUT: quantum advantage at the GUT scale
20th Patras Workshop on Axions, WIMPs, and WISPs, September 22-26, 2025, Tenerife, Spain

Status of DMRadio-50L and m^3
Identification of Dark Matter, July 18-22, 2022, Vienna, Austria

Completion of Phase I and Preparation for Phase II of the HAYSTAC Experiment
14th Patras Workshop on Axions, WIMPs, and WISPs, June 18-22, 2018, DESY, Hamburg, Germany

Teaching Experience

Head Teaching Assistant, Stanford University

Spr. 2022 Physics 25 – Modern Physics (Instructor: Kent Irwin).

2021-2022 **Mentor**, Polygence

One-on-one mentoring of high school students on research projects. Projects topics in dark matter physics and cosmology.

Teaching Assistant, Stanford University

Fall 2020 Physics 46 – Heat and Optics (Instructor: Giorgio Gratta).

Spr. 2020 Physics 43 – Electricity and Magnetism (Instructor: Mark Kasevich).